

EMCH 367 CONTROLS

FINAL EXAM PREVIEW

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December 5, 2024



ANNOUNCEMENT

- Final exam: December 10, Tuesday, 04:00 – 07:00 pm
- Course evaluation closes soon
 - More than 40 participants → Everyone gets extra credit
- All grades will be uploaded on Blackboard by December 13
 - All grades will be entered into the system by December 18
 - Deadline for appeal: December 18, 6:00 pm



FINAL EXAM PREVIEW

- P1: T/F ✓
- P2: Closed loop system ✓
 - (a) Closed loop transfer function ✓
 - (b) Partial fraction expansion ✓
 - (c) Routh stability ✓
- P3: PID ✓
 - (a) Stability analysis ✓
 - (b) Open loop ✓
 - (c) Root locus ✓
- P4: Bode/Nyquist ✓
- P5: Margins ✓
- BP: State space modeling of dynamic system ✓



REVIEW: LAPLACE TRANSFORMATION

- Enables mathematical modeling in s-domain
 - Time domain $\rightarrow$ s-domain
 - Differential equation $\rightarrow$ Algebraic equation
- Linear operator
- Final value theorem & Initial value theorem
- Partial fraction expansion



**Engineering
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REVIEW: TRANSFER FUNCTION

- Transfer function
 - Relationship between two variables in s-domain
- Block diagram
 - Graphical interpretation
 - Time domain & s-domain
- 1st order system
 - Time constant
- 2nd order system
 - Natural frequency
 - Damping ratio
- Stability of the system

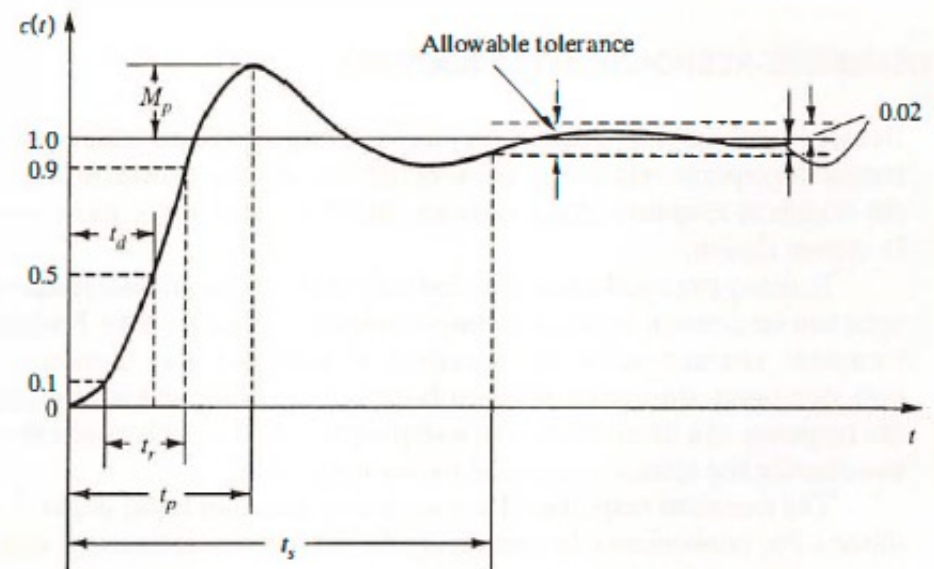
$$X(s) = \frac{1}{s^2 + \frac{b}{m}s + \frac{k}{m}} F(s) + \frac{s + \frac{b}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}} x(0) + \frac{1}{s^2 + \frac{b}{m}s + \frac{k}{m}} \dot{x}(0)$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



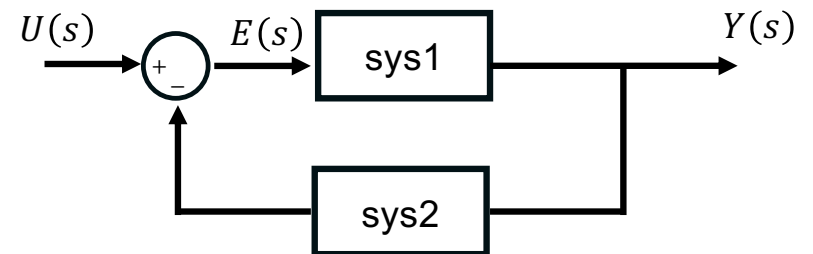
REVIEW: TIME RESPONSES

- Impulse / Step / Ramp
- Performance indicators
 - Steady state error
 - Rise time; Settling time; Peak time; Peak value; Max overshoot
- System ID



REVIEW: FEEDBACK SYSTEM

- Closed loop transfer function
- Block diagram of feedback system
- Types of the system
- System type



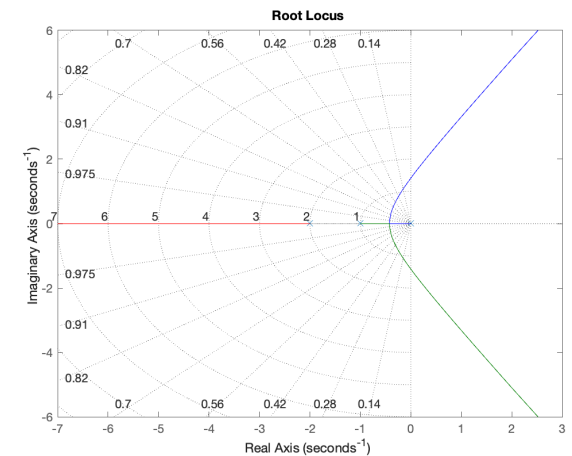
System		Steady state error	
Type	TF	Step error	Ramp error
0	$G(s) = K \frac{(T_a s + 1) \dots}{(T_1 s + 1) \dots}$		
1	$G(s) = \frac{K (T_a s + 1) \dots}{s (T_1 s + 1) \dots}$		
2	$G(s) = \frac{K (T_a s + 1) \dots}{s^2 (T_1 s + 1) \dots}$		



REVIEW: STABILITY

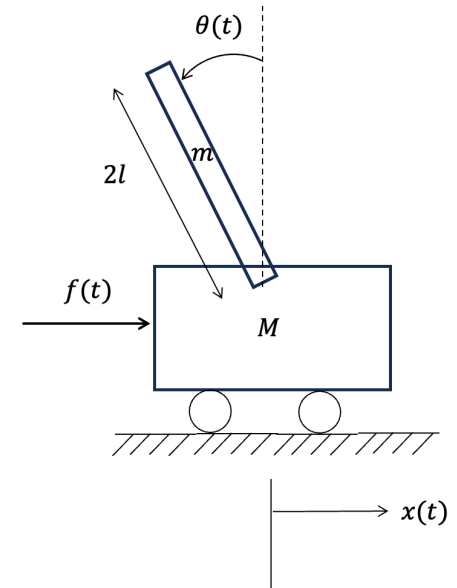
- Routh stability criterion
 - Stability of feedback system
- Root locus
 - Complex algebra
 - Magnitude & Phase
 - Steps to draw root locus
 - 2nd order system: natural frequency, damping ratio

s^n	a_0	a_2	a_4	a_6	$\dots$
s^{n-1}	a_1	a_3	a_5	a_7	$\dots$
s^{n-2}	b_1	b_2	b_3	b_4	$\dots$
s^{n-3}	c_1	c_2	c_3	c_4	
$\vdots$					
s^2	d_1	d_2			
s^1	e_1	e_2			
s^0	f_1				



REVIEW: PID CONTROL

- Inverted pendulum
- Proportional control
- Integral control
- Derivative control
- PI / PD / PID control



REVIEW: FREQUENCY-BASED ANALYSIS

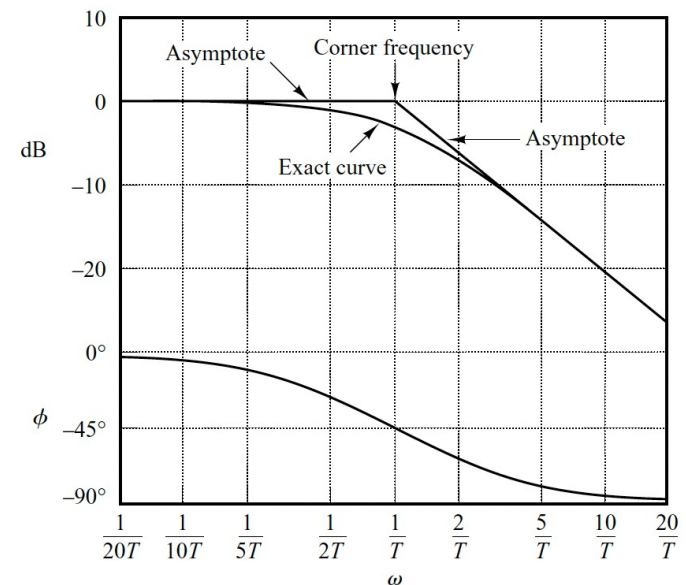
- Inputs are now sinusoidal

- $G(j\omega)$

$$y(t) = X|G(j\omega)| \sin(\omega t + \phi)$$

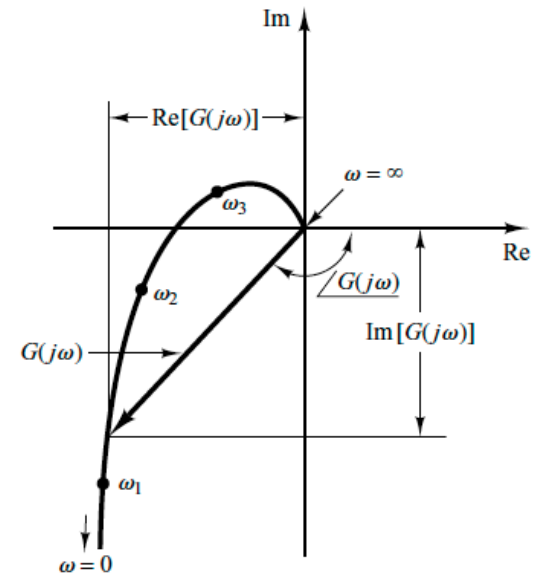
$$\phi = \angle G(j\omega) = \tan^{-1} \frac{\text{Im}(G(j\omega))}{\text{Re}(G(j\omega))}$$

- Bode diagram
 - Basic blocks
 - Resonant peak
 - Bandwidth



REVIEW: NYQUIST

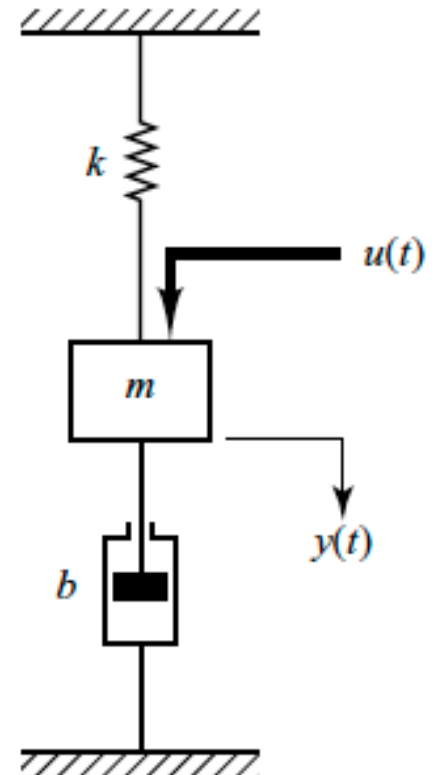
- Stability analysis in frequency domain
- Polar plots
- Mapping theorem
- Nyquist stability criterion
 - Nyquist path
 - Multiple sections
 - $Z = P + N$



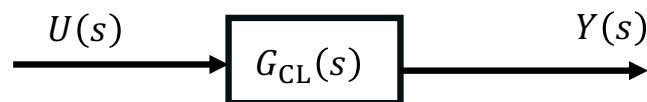
REVIEW: STATE SPACE MODELING

- What are the states?
- What are the advantages of state space modeling?
- Linear system

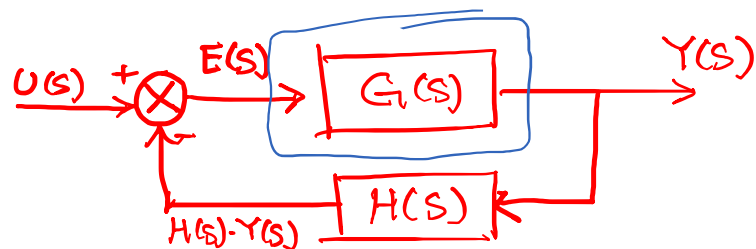
$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)u(t) \\ \mathbf{y}(t) &= \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)u(t)\end{aligned}$$



CLOSED LOOP TRANSFER FUNCTION



$$\frac{Y(s)}{U(s)} = G_{CL}(s) = \frac{G(s)}{1 + G(s)H(s)}$$



$$E(s) = U(s) - H(s) \cdot Y(s)$$

$$\begin{aligned} Y(s) &= G(s) \cdot E(s) \\ &= G(s) \cdot [U(s) - H(s) \cdot Y(s)] \\ &= G(s) \cdot U(s) - G(s) \cdot H(s) \cdot Y(s) \end{aligned}$$

$$Y(s)[1 + G(s) \cdot H(s)] = G(s) \cdot U(s)$$

$$\frac{Y(s)}{U(s)} = \frac{G(s)}{1 + G(s) \cdot H(s)}$$



EXAMPLE: PARTIAL FRACTION EXPANSION

- Find $y(t)$

$$Y(s) = \frac{5(s+2)}{s(s+1)(s+3)} = \frac{a_1}{s} + \frac{a_2}{s+1} + \frac{a_3}{s+3}$$

$$s \cdot Y(s) = a_1 + s \left[\frac{a_2}{s+1} + \frac{a_3}{s+3} \right] = \frac{5(s+2)}{(s+1)(s+3)}$$

$$s=0: a_1 = \frac{5 \cdot 2}{1 \cdot 3} = \frac{10}{3}$$

$$(s+1)Y(s) = a_2 + (s+1) \left[\frac{a_1}{s} + \frac{a_3}{s+3} \right] = \frac{5(s+2)}{s(s+3)}$$

$$s=-1: a_2 = \frac{5 \cdot 1}{(-1) \cdot 2} = -\frac{5}{2}$$

$$(s+3)Y(s) = a_3 + (s+3) \left[\frac{a_1}{s} + \frac{a_2}{s+1} \right] = \frac{5(s+2)}{s(s+1)}$$

$$s=-3: a_3 = \frac{5 \cdot (-1)}{(-3)(-2)} = -\frac{5}{6}$$

$$\mathcal{L}^{-1}\left[\frac{1}{s}\right] = 1(t)$$

$$\mathcal{L}^{-1}\left[\frac{1}{s+1}\right] = e^{-t}$$

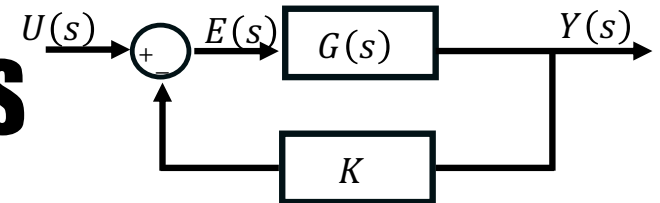
$$\mathcal{L}^{-1}\left[\frac{1}{s+3}\right] = e^{-3t}$$

$$\mathcal{L}^{-1}[Y(s)] = y(t)$$

$$= \frac{10}{3} \cdot 1(t) - \frac{5}{2} \cdot e^{-t} - \frac{5}{6} \cdot e^{-3t}$$



STABILITY OF FEEDBACK SYSTEMS



- Plant

$$G(s) = \frac{1-s}{s^2 + 2s + 4}$$

Closed loop transfer function

$$G_{CL}(s) = \frac{G(s)}{1 + KG(s)}$$

- Routh Stability

$$1 + KG(s) = 1 + K \cdot \frac{1-s}{s^2 + 2s + 4} = 0 \Rightarrow s^2 + (2-K)s + 4+K = 0$$

s^2	1	$4+K$
s^1	$2-K$	0
s^0	<u>$4+K$</u>	

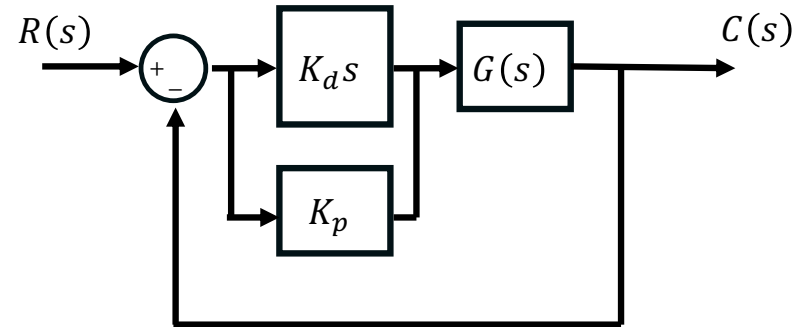
$$\frac{(2-K) \cdot (4+K) - 1 \cdot 0}{2-K} = 4+K > 0$$

$$\left. \begin{array}{l} 2-K > 0 \\ 4+K > 0 \end{array} \right\} -4 < K < 2$$



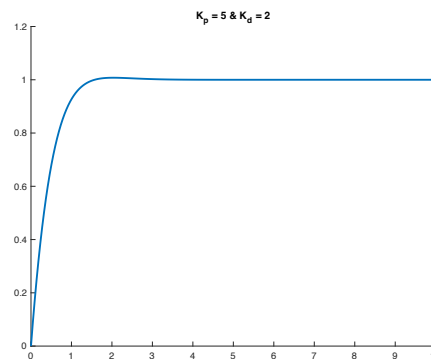
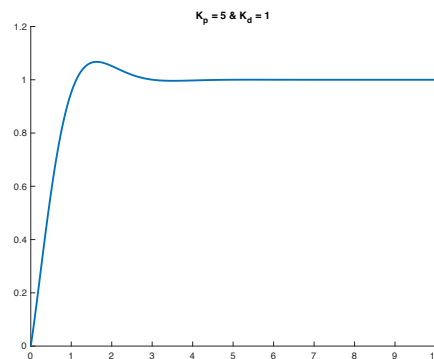
PD CONTROL

- Proportional & Derivative control



$$G_{CL}(s) = \frac{\frac{K_p + K_d s}{Js^2 + cs}}{1 + \frac{K_p + K_d s}{Js^2 + cs}} = \frac{K_d s + K_p}{Js^2 + (c + K_d)s + K_p}$$

- Step response



s^2 J
 s^1 $c + K_d$
 s^0 K_p



EXAMPLE: BODE DIAGRAM

$$\frac{1}{s} \rightarrow \frac{1}{j\omega}$$

$$|G(j\omega)| = \frac{1}{\omega}$$

$$20 \log |G(j\omega)| = -20 \log \omega$$

- Consider the transfer function

$$G(s) = \frac{10(s+3)}{s(s+2)(s^2+s+2)}$$

$$= \frac{10 \cdot 1.5 \left(\frac{s}{2} + 1\right)}{s \left(\frac{s}{2} + 1\right) \left(\frac{s^2}{2} + \frac{s}{2} + 1\right)}$$

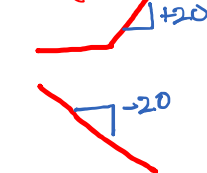
$$\frac{\frac{s}{3} + 1}{\frac{s}{2} + 1} \cdot \frac{1}{s} \cdot \frac{1}{\frac{s^2}{2} + \frac{s}{2} + 1}$$

$$\omega_c = 3$$

$$\omega_c = 2$$

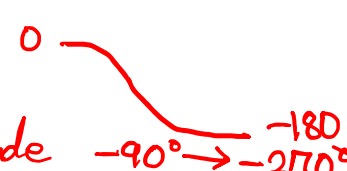
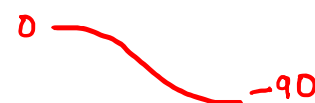
$$\omega_n = \sqrt{2}$$

Magnitude



-60 dB/decade

Phase



NYQUIST PATH

$$\frac{G}{1+GH}$$

$$0 = 1 + G(s)H(s) = \frac{s+1}{s-1}$$

- Nyquist path
 - The contour consists of the entire $j\omega$ axis from $\omega = -\infty$ to $\omega = \infty$ and a semicircular path of infinite radius in the RHP
 - It encloses the entire RHP $\rightarrow$ all zeros and poles of $1 + G(s)H(s)$ in RHP
 - If there are no zeros of $1 + G(s)H(s)$ in RHP, then there are no closed-loop poles in RHP, and the system is stable
 - Nyquist path should not pass through poles or zeros

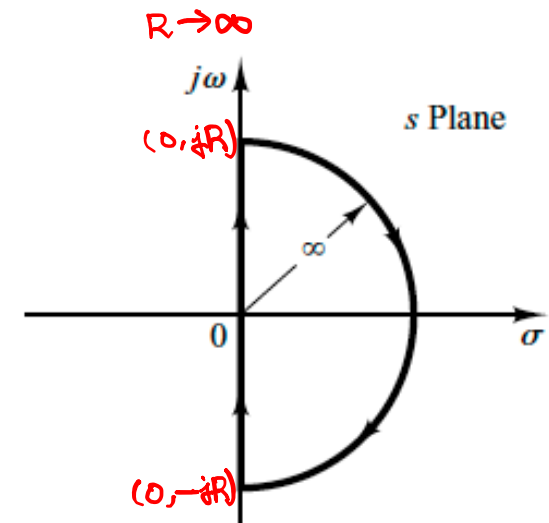
Mapping theorem

$$F(s) = 1 + G(s)H(s)$$

- The poles of $F(s)$ = The poles of $G(s)H(s)$
- For closed loop transfer function, the zeros of $1 + G(s)H(s)$ determines stability
- The number of zeros of $F(s) = 1 + G(s)H(s)$ in the RHP: Z

$$N = Z - P \Rightarrow \boxed{Z = N + P} \quad Z=0$$

origin of $1+GH$
 $\rightarrow -1$ of GH

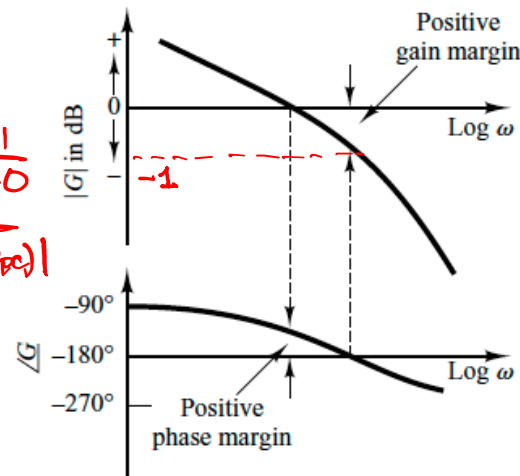


POSITIVE/NEGATIVE MARGINS

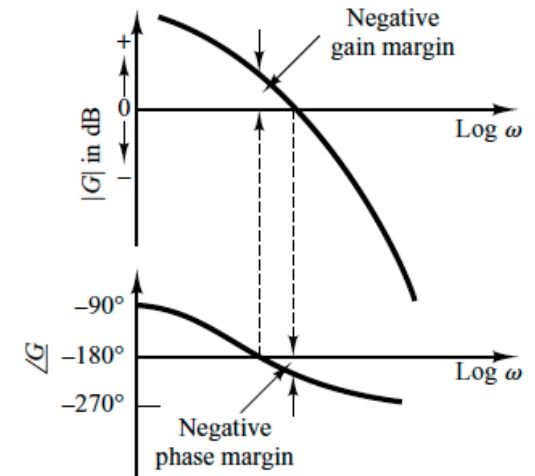
- Positive/negative phase margin
 - Sign of the phase margin
- Positive/negative gain margin
 - Bigger/less than 1

$$|G(j\omega_{pc})| = \frac{1}{10}$$

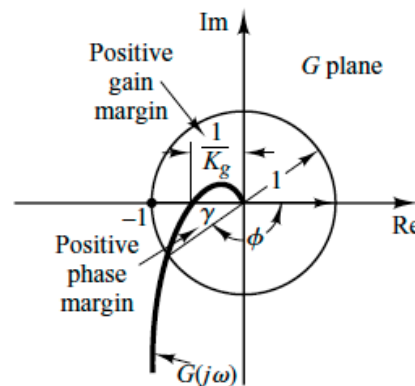
$$K_g = \frac{1}{|G(j\omega_{pc})|} = 10$$



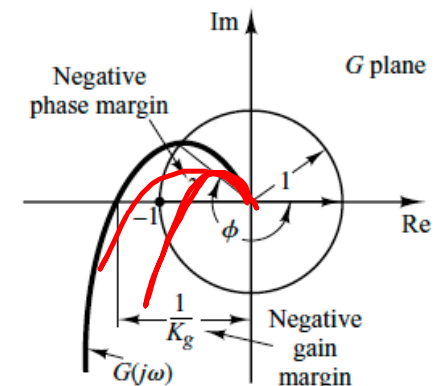
Stable system



Unstable system



Stable system



Unstable system